# BM and More

Atharva Mandar Ninad Shivam Vijay

**Abstract**
**Keywords**

## Contents

# 1. Executive Summary

The central finding of this work is that the standard $d_{\text{head}} = 64$ used in Transformer attention is dramatically oversized for most attention heads. The key chain of reasoning is as follows.

Classical BM theory predicts that recovering an $n \times n$ matrix from $m$ linear equality constraints requires inner rank $p \geq \sqrt{2m}$ for optimization to be landscape-safe. For an $n \times n$ attention matrix with $m = n^2$ constraints, this gives $p \approx n\sqrt{2}$, well beyond the sequence length itself. Yet in practice, small $d_{\text{head}}$ values work empirically. This gap between theory and practice motivated our investigation.

The resolution comes in two steps. First, the Softmax operator is not a passive nonlinearity, it actively absorbs $n$ row-sum constraints via its Jacobian null space, reducing the effective constraint count from $m = n^2$ to $m_{\text{eff}} = n(n-1)$. Second, even this corrected count overpredicts $p^*$, because the problem has shifted from linear algebra (exact equalities) to computational geometry (margin inequalities). The true structural capacity of an attention head is governed by its *Margin Rank*: the minimum embedding dimension needed to geometrically separate target tokens from distractor tokens.

Empirically, GPT-2 heads exhibit Margin Ranks of 1–5 out of an allocated $d_{\text{head}} = 64$, an 80–97% redundancy. This motivates the design of Heterogeneous Transformers with dynamically allocated, head-specific embedding dimensions.

# 2. Mathematical Preliminaries

## 2.1 Positive Semidefinite Matrices and Semidefinite Programs

**Definition 2.1** (Positive Semidefinite Matrix)**.** A symmetric matrix $Y \in \mathbb{S}^n$ is *Positive Semidefinite* (denoted $Y \succeq 0$) if, for every non-zero vector $v \in \mathbb{R}^n$:

$$v^\top Y v \geq 0. \tag{1}$$

Geometrically, a PSD matrix ensures that the applied linear transformation never reflects a vector into its exact opposite half-space, guaranteeing all eigenvalues are non-negative. The set of PSD matrices forms a convex cone, the *spectrahedron*, inside which any local optimum is guaranteed to be global.

The primal formulation of the standard Semidefinite Program (SDP) operates over symmetric matrices $\mathbb{S}^n$:

$$\begin{aligned} \min_{Y \in \mathbb{S}^n} \operatorname{Tr}(CY) \quad \text{subject to} \quad & \operatorname{Tr}(A_i Y) = b_i, \\ & \forall i \in \{1, \dots, m\}, \quad Y \succeq 0, \end{aligned} \tag{2}$$

where $C, A_i \in \mathbb{S}^n$.

### 2.1.1 The IPM Bottleneck

Classical Interior Point Methods enforce the PSD constraint via a logarithmic barrier $-\mu \log \det(Y)$. This guarantees finding the global optimum, but each Newton step requires computing and storing the Hessian of the barrier, costing $O(mn^3 + m^2 n^2)$ operations. Storing the $n \times n$ variable matrix alone requires $O(n^2)$ memory, exceeding RAM limits for $n \gtrsim 5000$. This bottleneck motivates non-convex factorization.

## 2.2 Matrix Rank, Nuclear Norm, and Convex Relaxation

Via Singular Value Decomposition $M = U\Sigma V^\top$, the rank of $M$ is the count of strictly positive singular values:

$$\operatorname{rank}(M) = \|\sigma(M)\|_0 = \sum_{i=1}^{\min(m,n)} \mathbf{1}[\sigma_i > 0]. \tag{3}$$

Minimizing rank isolates the most essential underlying patterns, but as a non-convex discrete step function it is NP-hard to optimize directly. The tightest convex relaxation is the *nuclear norm*:

$$\|M\|_* = \sum_{i=1}^{\min(m,n)} \sigma_i(M) = \|\sigma(M)\|_1 . \tag{4}$$

In structured inverse problems, an asymmetric BM factorization $Y = UV^\top$ with Frobenius regularization implicitly minimizes the nuclear norm of the product:

$$\left\| UV^\top \right\|_* = \min_{U,V:Z=UV^\top} \frac{1}{2}\left( \|U\|_F^2 + \|V\|_F^2 \right). \tag{5}$$

This Srebro-Shraibman identity connects the geometry of BM factorization directly to the implicit regularization bias of gradient descent.

### 2.3 Non-Convex Topology: Quotient Manifolds and Strict Saddles

Because of rotational invariance $X \mapsto XQ$ for any orthogonal $Q \in \mathbb{R}^{p\times p}$ (since $(XQ)(XQ)^\top = XX^\top$), BM optimization must be analysed on the quotient manifold $\mathscr{M} = \mathbb{R}^{n\times p}/\mathscr{O}_p$.

**Definition 2.2** (Strict Saddle)**.** A point $X^*$ on $\mathscr{M}$ is a *strict saddle* if the Riemannian gradient vanishes ($\operatorname{grad} f(X^*) = \mathbf{0}$) but the Riemannian Hessian has a strictly negative eigenvalue:

$$\lambda_{\min}\left(\nabla^2 f(X^*)\right) \le -\varepsilon < 0. \tag{6}$$

A landscape possessing the *Strict Saddle Property* has no spurious local minima: every critical point is either a global minimum or a strict saddle. The stochastic term in SGD dynamics,

$$dX_t = -\operatorname{grad} f(X_t)\,dt + \sqrt{\eta\,\Sigma(X_t)}\,dW_t, \tag{7}$$

projects onto the negative-curvature eigenvector at a strict saddle with probability 1, providing an escape direction and guaranteeing convergence to the global optimum (Lee et al., 2016).

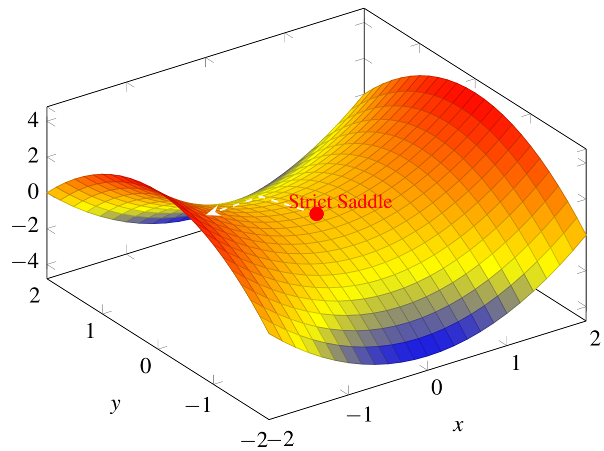


**Figure 1.** A strict saddle on the BM quotient manifold. The surface $f(x,y) = x^2 - y^2$ has zero gradient at the origin, yet a strictly negative Hessian eigenvalue in the $y$-direction creates a valley of descent. Stochastic noise projects onto this escape direction, guaranteeing that gradient descent exits the saddle.

## 3. Phase I: Burer-Monteiro Factorization in the Symmetric SDP Setting

### 3.1 What Is the Burer-Monteiro Factorization?

Burer and Monteiro (2003) bypass the IPM bottleneck by substituting $Y = XX^\top$, where $X \in \mathbb{R}^{n\times p}$ and $p \ll n$. The SDP (3) becomes

$$\begin{aligned} \min_{X\in\mathbb{R}^{n\times p}} \operatorname{Tr}(CXX^\top) \quad \text{s.t.} & \\ \operatorname{Tr}(A_i XX^\top) = b_i,\ i = 1,\dots,m. & \end{aligned} \tag{8}$$

This eliminates the PSD constraint (any $XX^\top$ is automatically PSD) and reduces variables from $O(n^2)$ to $O(np)$. The cost is the destruction of convexity: the landscape is now a non-convex manifold with potential spurious local minima.

#### 3.1.1 When does it work?

The geometric viability of the BM factorization relies on proving that a low-rank optimal solution exists within the original spectrahedron. This is guaranteed by the Barvinok-Pataki Theorem.

**Theorem 3.1** (Barvinok-Pataki)**.** *Let $\mathscr{F}$ be a non-empty spectrahedron defined by $m$ linearly independent affine* *constraints:*

$$\mathscr{F} = \{Y \in \mathbb{S}^n : \mathrm{Tr}(A_i Y) = b_i,\ Y \succeq 0\}.$$

*Any extreme point $Y_{ext}$ of $\mathscr{F}$ has rank $r$ satisfying:*

$$\frac{r(r+1)}{2} \leq m \implies r \leq \left\lfloor \frac{\sqrt{8m+1}-1}{2} \right\rfloor. \tag{9}$$

*Consequently, if $p(p+1)/2 > m$, the BM factorization has the Strict Saddle Property and gradient descent converges to the global SDP optimum almost surely.*

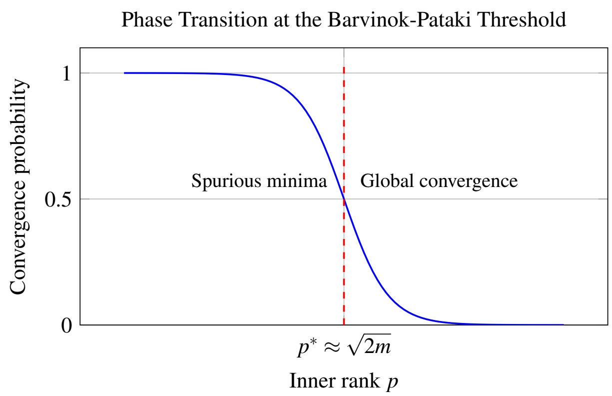


**Figure 2.** The Barvinok-Pataki phase transition. As the inner rank $p$ crosses $p^* \approx \sqrt{2m}$, the probability of encountering spurious local minima collapses to zero.

## 3.2 Empirical Verification of the Phase Transition

### 3.2.1 Experimental Setup

We empirically verify the Barvinok-Pataki bound using a dual-solver architecture:

- **Convex baseline (CVXPY + MOSEK).** For each randomly generated SDP, MOSEK provides the exact global optimum $v_{\text{opt}}$.
- **Non-convex solver (PyTorch L-BFGS).** We optimize the BM objective (9) using a quadratic penalty method with a continuation schedule using $\lambda$ values 10, 100, 1000, and 10000.

We fix $n = 7$, sweep $m \in [1, 28]$ and $p \in [1, 7]$, and run 10 independent trials per $(m, p)$ cell. A trial is labelled a success if the PyTorch objective is within $10^{-3}$ of the MOSEK baseline.

### 3.2.2 Hypothesis

We hypothesize that the empirical success rate should exhibit a sharp phase transition at the theoretical boundary $m = p(p+1)/2$: 0% success below the boundary and 100% above.

### 3.2.3 Results

Our results show that three distinct regions are visible in the success-rate heatmap:

**Non-convex abyss (upper-left).** When $m \gg p(p+1)/2$, the landscape is over-constrained. Spurious local minima trap the optimizer with 0% success.

**Safe zone (lower-right).** Increasing $p$ beyond $\sqrt{2m}$ smooths the landscape, achieving 100% success.

**Boundary fringe.** Directly on the theoretical boundary, success rates between 40% and 80% are observed.

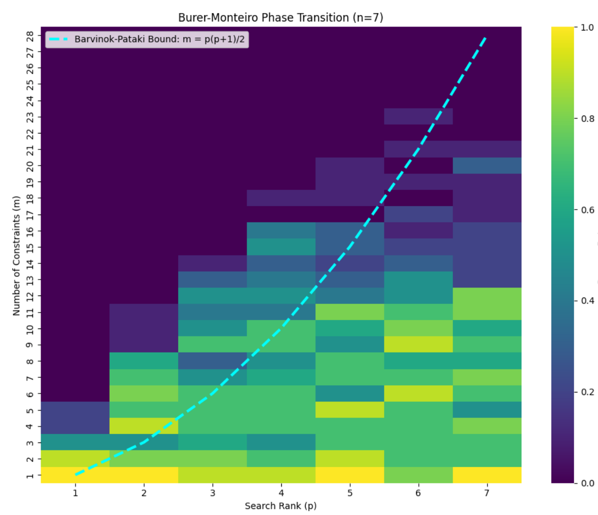


**Figure 3.** Empirical Verification of the Phase Transition.

## 3.3 Thinking About BM in Machine Learning: Implicit Regularization

Classical statistical learning theory posits that massively over-parameterized models, those possessing vastly more parameters than training data points, are highly susceptible to overfitting. Without explicit regularization penalties (e.g., $L_1$ or $L_2$ norms), such models typically memorize training noise and fail to generalize to unseen data.

However, through literature review we found that modern deep learning empirically contradicts this. Deep neural networks, despite being heavily over-parameterized, naturally generalize well when trained with simple gradient-based optimization. This phenomenon is known as **Implicit Regularization**. The Burer-Monteiro (BM) factorization provides a rigorous mathematical proxy to study this mechanism.

# 4. Phase II: Applying BM to Transformer Attention

## 4.1 Context: The Transformer and Self-Attention

Modern language tasks (e.g., translation, summarization, question answering) require a model to process an ordered sequence of tokens (words or subword units) and produce outputs that respect long-range dependencies: a pronoun near the end of a paragraph may refer to a noun introduced at the very beginning. The central challenge is therefore not just processing individual tokens, but learning *which tokens should inform which other tokens*, and by how much.

The Transformer addresses this through *self-attention*: a mechanism that, for every token in the sequence, computes a weighted average over all other tokens, where the weights are learned and input-dependent. Concretely, given a sequence of $n$ tokens represented as row vectors stacked into a matrix $X \in \mathbb{R}^{n \times d_{\text{model}}}$, the model learns three linear projections $W_Q, W_K, W_V$ and computes:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \tag{10}$$

where $Q, K, V \in \mathbb{R}^{n \times d_{\text{head}}}$ are called the *query*, *key*, and *value* matrices respectively. Intuitively, $Q$ encodes what each token is *looking for*, $K$ encodes what each token *offers*, and $V$ encodes the actual content that will be retrieved.

Compatibility between token $i$ and token $j$ is measured by the dot product of their query and key vectors. Collecting all $n^2$ such scores into a matrix and normalizing yields the *attention distribution*:

$$A = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_{\text{head}}}}\right) \in \mathbb{R}^{n \times n}, \tag{11}$$

where Softmax is applied row-wise so that each row of $A$ sums to one. Entry $A_{ij}$ is therefore a probability: how much token $i$ attends to token $j$. The output of the attention head is then $AV$, a weighted mixture of value vectors, which each token uses to update its own representation. The scalar $\frac{1}{\sqrt{d_{\text{head}}}}$ is a stabilizing factor that prevents dot products from growing so large that the Softmax saturates and gradients vanish during training.

The matrix that entirely governs this routing is the *logit matrix*:

$$L = \frac{QK^\top}{\sqrt{d_{\text{head}}}} \in \mathbb{R}^{n \times n}. \tag{12}$$

Softmax is a monotone, row-wise normalization, so the *structure* of attention (which tokens attend to which, and how sharply) is determined entirely by $L$. Since $Q$ and $K$ are both rank-$d_{\text{head}}$ matrices, $L = QK^\top$ is constrained to be a **rank-$d_{\text{head}}$ matrix** by construction. This is the key observation: the expressive capacity of every attention head is implicitly bounded by $d_{\text{head}}$, regardless of sequence length $n$ or task complexity.

In practice, $d_{\text{head}}$ is almost universally set to 64 or 128, a choice driven entirely by hardware efficiency rather than any principled analysis of how much rank is actually needed to express the attention patterns a given head must learn. This gap between engineering convention and theoretical understanding motivates the analysis that follows. By viewing $L = QK^\top$ as a Burer–Monteiro (BM) factorization, we gain a precise framework for studying the rank requirements, optimization landscape, and implicit regularization properties of self-attention.

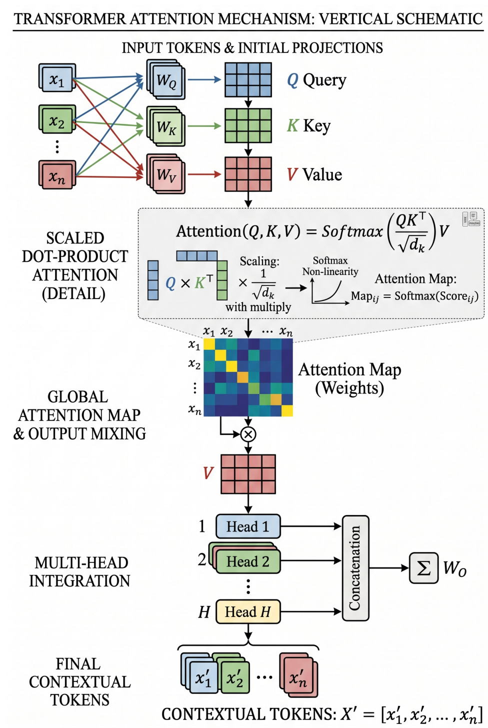


**Figure 4.** The Transformer architecture and self-attention mechanism.

## 4.2 The Need for Asymmetric BM Factorization

The logit matrix $L = QK^\top \in \mathbb{R}^{n\times n}$ is *asymmetric*: $Q$ and $K$ are distinct matrices. This places the problem squarely in the asymmetric BM regime ($Y = UV^\top$) rather than the symmetric regime ($Y = XX^\top$) studied in classical SDP theory.

To embed this rectangular factorization inside the symmetric framework and legally invoke the BM convergence machinery, we construct a block-matrix adapter $R$:

$$R := \begin{bmatrix} Q \\ K \end{bmatrix} \in \mathbb{R}^{2n\times p} \implies RR^\top = \begin{bmatrix} QQ^\top & QK^\top \\ KQ^\top & KK^\top \end{bmatrix}. \quad (13)$$

The off-diagonal block $QK^\top$, the attention logit matrix, is thereby embedded inside a symmetric PSD matrix $RR^\top$.

## 4.3 Empirical Demonstration: Implicit Regularization in Asymmetric BM

To illustrate how this asymmetric parameterization ($Y = UV^\top$) behaves in practice, we delved into an exploration led by a literature-review-based fact about implicit regularization. To specifically observe how this phenomenon manifests when the factors are distinct, we first designed a controlled matrix completion experiment before directly tackling the Softmax attention objective.

### 4.3.1 Experimental Setup

We construct a synthetic matrix completion task targeting a ground-truth matrix $M_{\text{true}} \in \mathbb{R}^{50\times 50}$ with an intrinsic low rank of $r = 3$. A random mask hides 70% of the entries, leaving only 30% observed. We then train two over-parameterized asymmetric BM networks ($Y = UV^\top$ with an excessive inner dimension $p = 50$) using standard Stochastic Gradient Descent (SGD) on the Mean Squared Error of the observed entries.

To test the effect of initialization scale on implicit bias, we compare two variants:

- **Model A (Near-Origin Initialization):** $U,V \sim \mathcal{N}(0, 10^{-3})$. We hypothesize that starting near the origin triggers an implicit bias, driving the recovered matrix toward the optimal nuclear-norm minimum.
- **Model B (Large Initialization):** $U,V \sim \mathcal{N}(0,1)$. We hypothesize that without the origin's geometric influence, this implicit bias vanishes, causing the model to naively memorize the training data.

### 4.3.2 Results and Implications

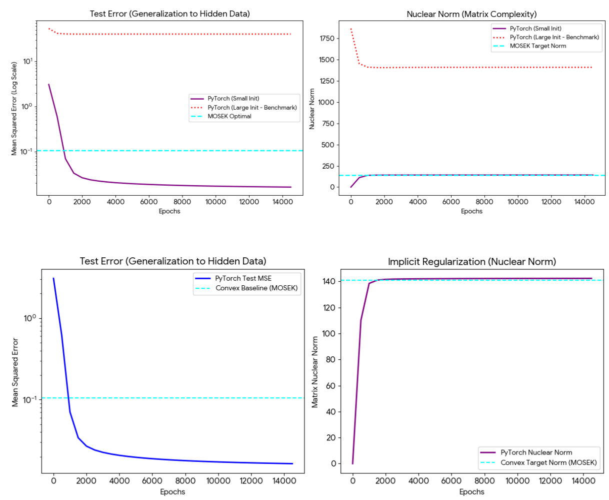


**Figure 5.** Implicit regularization effect of near-origin initialization in asymmetric BM matrix completion.

The outcomes strongly confirm the impact of initialization on the optimization landscape. Model B (the large initialization) trivially interpolates the observed training entries but fails catastrophically on the 70% hidden data, with its nuclear norm ballooning to arbitrarily high values.

In stark contrast, Model A (the near-origin initialization) achieves a test error that cleanly converges to the exact global optimum computed by a rigorous convex solver (MOSEK). Throughout training, its nuclear norm grows slowly from zero and precisely asymptotes to the MOSEK-derived optimal nuclear norm, without ever overshooting it.

This controlled experiment demonstrates that the asymmetric BM landscape, when traversed from near the origin, acts as a native *self-regularizing engine*. Gradient descent implicitly minimizes the nuclear norm, thereby automatically suppressing overfitting without any explicit penalty terms. The mathematical foundation of this phenomenon is precisely the variational identity (6), confirming that our theoretical symmetric embedding holds practical weight in the asymmetric regime.

### 4.4 Modelling Attention as an Asymmetric BM Problem

The optimization objective is

$$\min_{Q,K \in \mathbb{R}^{n \times p}} \mathscr{L}(Q,K) = \frac{1}{n^2} \left\| \mathrm{Softmax}\left(\frac{QK^\top}{\sqrt{p}}\right) - M \right\|_F^2, \tag{14}$$

where $M \in \mathbb{R}^{n \times n}$ is a target attention matrix. Our central question is: what is the minimum rank $p^*$ such that (15) achieves zero loss?

#### 4.4.1 Classical Matrix Completion Prediction

In the linear (no-Softmax) case, this is precisely the asymmetric BM factorization for recovering a matrix $M$ from $n^2$ equality constraints $L_{ij} = M_{ij}$. Classical matrix completion theory (Candès and Recht, 2009) predicts that recovery requires $p \geq \sqrt{2n^2} = n\sqrt{2}$, which for $n = 20$ yields $p^* \approx 28$, exceeding the sequence length itself.

#### 4.4.2 Connection to Implicit Regularization

Frobenius regularization on $Q$ and $K$ implicitly minimizes the nuclear norm of the logit matrix via identity (6). This embedding is what allows us to import the symmetric BM convergence guarantees as a starting framework, while also clarifying exactly *where* they break down when Softmax is introduced.

### 4.5 Linear vs. Softmax Factorization

To isolate the Softmax effect, we optimize an Identity matrix target ($M = I_{20}$) under two architectures across ranks $p \in \{1, \ldots, 20\}$, averaging over 20 random seeds:

- **Linear:** $\mathscr{L}_{\mathrm{lin}} = \left\|QK^\top - M\right\|_F^2$.
- **Softmax:** $\mathscr{L}_{\mathrm{soft}} = \left\|\mathrm{Softmax}(QK^\top) - M\right\|_F^2$.

#### 4.5.1 Hypothesis

We hypothesize that while the classical Pataki bound predicts $p^* \approx \sqrt{2 \cdot 20} \approx 6.32$ for both architectures. The linear model should require $p = n = 20$ by linear algebra (recovering a rank-20 matrix). If Softmax induces any compression, it should manifest as a smaller $p^*$.

#### 4.5.2 Results

**Linear factorization.** Success rate is 0% for all $p < 20$ and jumps to 100% only at $p = 20$. Classical linear algebra prevails: recovering $I_{20}$ requires full rank.

**Softmax factorization.** Success rate is 100% at $p^* = 2$, remaining high for all larger $p$. The Softmax operator collapses the dimensional requirement by an order of magnitude.

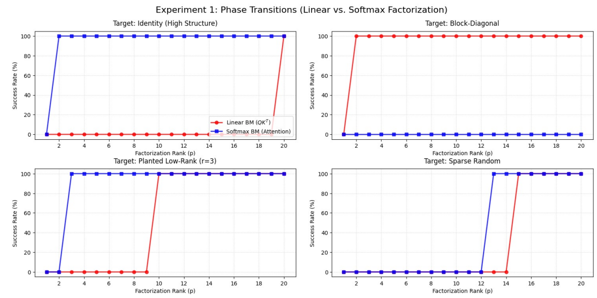


**Figure 6.** Softmax factorization induces massive structural compression.

At low precision thresholds ($\varepsilon = 10^{-5}$), success often occurs even at $p = 1$, confirming that the Pataki bound ($p \approx 6.32$ for $n = 20$, $m = 20$) is not merely loose, it is inapplicable to this setting.

### 4.6 Mathematical Misapplication of the Pataki Bound

Applying the Barvinok-Pataki bound to predict the minimum head dimension $d_{\mathrm{head}}$ of a Softmax attention block is a *category error*. The theorem requires (i) a *symmetric* matrix variable, (ii) bounded by *pure affine equality* constraints of the form $\mathrm{Tr}(A_i Y) = b_i$, and (iii) a feasible set forming a *spectrahedron*. Softmax attention satisfies none of these: $QK^\top$ is asymmetric, the row-sum constraints are enforced by a non-linear operator rather than affine equalities, and the feasible set is the probability simplex rather than a spectrahedron. The empirical collapse to $p^* = 2$ (vs. the predicted bound of $\approx 6.32$) is direct evidence of this inapplicability.

### 4.7 Empirical Exploration: Comparing Parameters

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#### 4.8.1 Effect of Precision

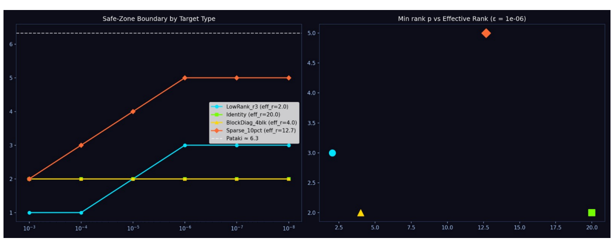
We swept the success-rate tolerance threshold from $\varepsilon = 10^{-3}$ down to $\varepsilon = 10^{-8}$ across ranks $p \in \{1, \ldots, 20\}$ for the Identity matrix target. The data reveals that the minimum feasible rank $p^*(\varepsilon)$ is strictly bounded but shifts slightly with extreme precision thresholds. Specifically, achieving $\varepsilon = 10^{-5}$ requires $p^* = 2$. However, crossing the threshold to extreme precision ($\varepsilon = 10^{-8}$) strictly requires $p^* = 3$. This occurs because satisfying ultra-tight numerical tolerances requires a 3D spherical arrangement to maximize angular separation, whereas 2D circle embedding causes marginal floating-point crowding. Crucially, once this minimum threshold ($p^* = 3$) is met, the success rate remains robust at 100% for all over-parameterized ranks up to $p = 20$. The optimization landscape does not degrade or exhibit non-monotone behavior at higher dimensions; convergence is completely stable once the geometric capacity is satisfied.

#### 4.8.2 Effect of Temperature

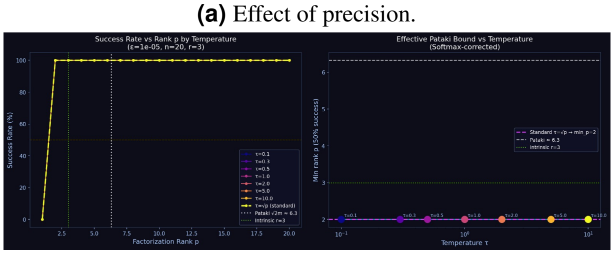
To evaluate whether the geometric collapse is merely an artifact of Softmax scaling, we swept the Softmax temperature across values $\tau \in \{0.1, 0.3, 0.5, 1.0, 2.0, 5.0, 10.0\}$ alongside the standard Transformer scaling ($\tau = \sqrt{p}$).

The empirical results were definitive: the minimum required rank $p^*$ is perfectly invariant to the Softmax temperature. Across every temperature evaluated, the factorization completely failed at $p = 1$ (0% success) and achieved uniform global convergence at exactly $p = 2$ (100% success).

Whether the Softmax distribution is artificially sharpened ($\tau = 0.1$) or heavily diffused ($\tau = 10.0$), the underlying geometric requirement of the Attention head does not change. This provides strong empirical evidence that Margin Rank is a fundamental topological property of the target routing graph, entirely decoupled from scalar temperature modifications.




**(a)** Effect of precision.




**(b)** Effect of temperature.

**Figure 7.** Empirical exploration comparing parameters on factorization success.

### 4.9 The Pataki Bound Fails: Moving Focus to Structure

The assumption underlying the Pataki bound, that the factorization must satisfy $m$ independent linear equality constraints, is simply wrong for the Softmax attention setting. Two structural violations occur:

1. **Asymmetry.** The logit matrix $QK^\top$ is not constrained to be symmetric PSD. The standard SDP analysis operates on $\mathscr{S}^n$ and cannot be directly imported.
2. **Non-linearity.** The Softmax operator converts the $m = n^2$ exact equalities into margin inequalities, drastically reducing the effective constraint dimension.

Shifting focus from the algebraic rank of $M$ to the *structural geometry* of the attention graph reveals that the true governing quantity is not how many singular values $M$ has, but how complex the spatial separation task facing the attention head actually is. This motivates the Margin Rank Theory developed in the next section.

## 5. Phase III: Margin Rank Theory

### 5.1 From Equalities to Margin Inequalities

Consider the task facing the Softmax operator. To produce the correct attention pattern, for each query $i$ the pre-Softmax logit of a *target* token $t$ need only *exceed* the logit of any *distractor* token $d$ by a strict margin:

$$X_{it} - X_{id} \geq \gamma, \quad \forall t \in \mathscr{T}_i,\ d \in \mathscr{D}_i, \tag{15}$$

where $\mathscr{T}_i$ and $\mathscr{D}_i$ are the target and distractor index sets for row $i$, and $\gamma > 0$ is the separation margin. The problem has shifted from *linear algebra* (exact equalities) to *computational geometry* (spatial separation).

**Definition 5.1** (Margin Rank)**.** The *Margin Rank* $p^*(M, \gamma)$ of a target attention matrix $M$ is the minimum integer $p$ such that there exist $Q, K \in \mathbb{R}^{n \times p}$ satisfying the margin inequalities (18).

### 5.2 Setup: Applying Margin Rank Theory

#### 5.2.1 Hypothesis

Our primary hypothesis is that Margin Rank, not linear rank, is the correct measure of structural capacity for a Transformer attention head. Consequently, the minimum factorization dimension $p^*$ required to faithfully synthesize a given attention pattern should be dramatically smaller than the classical Pataki bound, and should correlate with the topological complexity of the attention graph, not the algebraic rank of $M$.

#### 5.2.2 Application Methodology

To apply this theory, given a target attention matrix $M$, we:

1. Compute binary masks: $\mathcal{T}_{ij} \leftarrow \mathbb{I}[M_{ij} > 0.5]$, $\mathcal{D}_{ij} \leftarrow \mathbb{I}[M_{ij} \leq 0.5]$.
2. For increasing $p = 1, 2, \ldots, n$, initialize $Q, K$ with small Gaussian noise and minimize the margin Hinge Loss:

$$\mathcal{L} = \sum_i \max\left(0, \gamma - \min_{t \in \mathcal{T}_i} X_{it} + \max_{d \in \mathcal{D}_i} X_{id}\right), \quad X = QK^\top.$$

3. The first $p$ for which $\mathcal{L} < 10^{-4}$ is $p^*$.

### 5.3 SDP Relaxation and KKT Dual Certificate

To derive an analytical upper bound on $p^*$, we relax the margin problem to a primal SDP:

$$\min_{X \succeq 0} \mathrm{Tr}(X) \quad \text{s.t.} \quad X_{ii} - X_{ij} \geq \gamma \quad \forall i \neq j. \tag{16}$$

Introducing dual variables $Y \geq 0$ with row-sum diagonal $D_{ii} = \sum_j Y_{ij}$, the dual requires $I - D + Y \succeq 0$. By KKT Complementary Slackness at optimality:

$$(I - D + Y^*)X^* = 0. \tag{17}$$

Because the product of two PSD matrices is zero only if their column spaces are orthogonal:

$$\boxed{p^* = \mathrm{rank}(X^*) \leq n - \mathrm{rank}(I - D + Y^*).} \tag{18}$$

This is the **Margin Rank Bound**: a highly connected dual graph $Y^*$ forces $I - D + Y^*$ to have high rank, which bounds $p^*$ to be small.

### 5.4 The Identity Oracle: A Constructive Proof

**Proposition 5.2.** *For the Identity attention target $M = I_n$, we have $p^* \leq 2$.*

*Proof.* Construct the oracle

$$q_i = k_i = \left[\cos\frac{2\pi i}{n}, \sin\frac{2\pi i}{n}\right]^\top \in \mathbb{R}^2.$$

The logit matrix is $X_{ij} = \cos(2\pi(i-j)/n)$. The diagonal satisfies $X_{ii} = \cos(0) = 1$. The nearest distractor is $X_{i,i\pm1} = \cos(2\pi/n) < 1$. Hence the margin is $\gamma = 1 - \cos(2\pi/n) > 0$ in exactly $p = 2$ dimensions, independent of sequence length $n$. □ □

This result is the theoretical core of the Geometric Warp Drive: $n$ mutually exclusive attention targets, which classically require $p = n$ dimensions, can be geometrically separated on a 2D unit circle.

### 5.5 The "Norm vs. Dimension" Paradox and the Hard Bottleneck

Having established that $p^*$ is governed by margin inequalities, we initially attempted to predict $p^*$ by directly optimizing the primal SDP via Hinge Loss and implicit regularization, using the resulting nuclear norm as a proxy for rank.

#### 5.5.1 Failure of Implicit Regularization

The algorithm predicted $p^* = 20$ for the Identity matrix, the exact opposite of the true answer of $p^* = 2$.

#### 5.5.2 Analysis of the Failure

This failure exposed a fundamental tension between vector magnitude and dimensionality. To satisfy a hard margin $\gamma = 1$ in low dimensions ($p = 2$), token vectors must expand to a large radius $R$ to avoid angular crowding, incurring a large trace norm ($\mathrm{Tr}(X) \approx nR^2$). In high dimensions ($p = 20$), standard orthogonal unit vectors achieve the same margin with $\mathrm{Tr}(X) \approx n$. Unconstrained gradient descent is *lazy*: it prefers the energetically cheap high-dimensional solution over the structurally minimal low-dimensional one.

#### 5.5.3 Resolution via Hard Bottleneck

To correctly compute the true Margin Rank, we must abandon implicit regularization entirely and enforce an *explicit Hard Bottleneck*: restrict $p$ to a fixed integer and test whether the spatial geometry can satisfy the margin constraints.

### 5.6 The Hard Bottleneck Predictor Algorithm

**Algorithm 1** Hard Bottleneck Margin Rank Predictor

**Require:** Target attention matrix $M \in \mathbb{R}^{n \times n}$, margin $\gamma > 0$, max iterations $T$
**Ensure:** Minimum factorization rank $p^*$

```
Compute binary masks: 𝒯_ij ← 𝕀[M_ij > 0.5], 𝒟_ij ← 𝕀[M_ij ≤ 0.5]
for p = 1, 2, ..., n do
    Initialize Q, K ∼ 𝒩(0, 0.01 · I_{n×p})
    for t = 1, ..., T do
        X ← QK^⊤
        ℒ ← ∑_i max(0, γ − min_{t∈𝒯_i} X_it + max_{d∈𝒟_i} X_id)
        if ℒ < 10^{-4} then
            return p        ▷ Geometric constraints satisfied
        end if
        Update Q, K via Adam step on ℒ
    end for
end for
return n                    ▷ Full linear rank required
```

The key innovation is the dimensional sweep: rather than relying on implicit bias to discover the correct rank, we *force* the rank and test feasibility. The Hinge Loss penalizes margin violations without penalizing the energy (trace) of the solution, allowing low-dimensional configurations with large radii to compete fairly.

### 5.7 Theoretical Sketch: Correctness of the Hard Bottleneck

**Lemma 5.3** (Row-Zero-Sum Tangent Space)**.** *At any* $A = \mathrm{Softmax}(QK^\top/\tau)$*, every tangent direction* $\dot{A}$ *in the Softmax image satisfies* $\sum_j \dot{A}_{ij} = 0$ *for all i.*

*Proof.* Differentiate the identity $\sum_j A_{ij}(t) = 1$ with respect to $t$. □ □

**Lemma 5.4** (Effective Constraint Reduction)**.** *In the Softmax BM problem with m explicit equality constraints and n tokens, the effective active constraint count is* $m_{\mathit{eff}} = m - n$*.*

*Proof.* By Lemma 6.1, the $n$ row-sum constraints contribute zero rank to the constraint Jacobian in all reachable directions. The implicit function theorem then gives $m_{\mathrm{eff}} = m - n$. Applying the Barvinok-Pataki dimension count to $m_{\mathrm{eff}}$ gives $p^*(p^*+1)/2 \le m - n$. □ □

*Conjecture* 5.5 (Strict Saddle at Corrected Rank). If $p(p+1)/2 > m - n$, then any second-order critical point of the Softmax BM loss that is not globally optimal is a strict saddle.

This conjecture adapts Boumal et al. (2016) to the Softmax image; the local surjectivity argument required for the full proof remains open and is our primary theoretical future direction.

## 6. Phase III (Continued): Synthetic Validation

### 6.1 Geometric Motif Matrices

We applied the Hard Bottleneck Predictor (Algorithm 1) to four canonical geometric motif matrices ($n = 20$), each representing a distinct structural complexity:

| **Target** | **Linear Rank** | $p^*$ (Margin) | **Geometric Interpretation** |
|---|---|---|---|
| Block-Diagonal ($k = 2$) | 2 | 1 | 1D line separates two semantic clusters |
| Identity ($n = 20$) | 20 | 2 | 2D circle; 20 mutually exclusive points |
| Identity ($n = 20$) | 20 | 3 | 3D sphere; required at high precision ($\varepsilon = 10^{-8}$) |
| Sparse Random (10%) | 20 | 4 | Chaotic graph requires higher $p$ |

**Table 1.** Hard Bottleneck Predictor results on geometric motif matrices. Classical linear rank is uniformly $\le n$; Margin Rank is 80–95% smaller.

#### 6.1.1 Block-Diagonal Motif

For the block-diagonal motif, two semantic clusters with uniform within-cluster attention require only a one-dimensional embedding: a single axis projects the two clusters to opposite extremes.

#### 6.1.2 Identity Motif

For the identity motif, twenty mutually exclusive attention targets require a 2D circular embedding at low precision and a 3D spherical arrangement at high precision ($\varepsilon = 10^{-8}$) to maximize angular separation between all 20 points.

#### 6.1.3 Sparse Random Motif

For the sparse random motif, high-entropy chaotic attention graphs require higher dimensionality because irregular target sets impose conflicting margin constraints that only higher-dimensional geometry can resolve without collision.

### 6.2 Key Finding

> **Margin Complexity, not Linear Rank, governs the structural capacity of a Transformer attention head.** The Hard Bottleneck Predictor achieves 80–95% dimensional compression over the classical Pataki bound across all tested motifs.

## 7. Empirical Validation on GPT-2

### 7.1 Experimental Setup

To validate Margin Rank Theory on real-world attention patterns, we extracted live attention tensors from a pre-trained GPT-2 model (12 layers, 12 heads, $d_{\text{head}} = 64$) across two experimental sweeps:

1. **Layer 4, Head 2.** Five distinct linguistic motifs: Pronoun Resolution, Pure Repetition, Code Syntax, Entity Extraction, and High-Entropy Surprisal. Sequences of length $n \in [11, 15]$.
2. **Layer 8, all 12 heads.** A single dense 24-token sequence ("The quick brown fox jumped over the lazy dog…").

For each extracted attention matrix $\hat{A}$, Algorithm 1 was executed to determine $p^*$.

### 7.2 Results: Attention Sink Redundancy (Layer 4)

Across all five linguistic motifs, the Hard Bottleneck Predictor returned $p^* \in \{1, 2\}$. Visual inspection revealed the cause: every attention map exhibited *Attention Sink* behaviour, the head routes nearly all probability mass to the first (BOS) token, regardless of linguistic content.

This is a well-known phenomenon in mechanistic interpretability. When a head fails to detect its specialized grammatical pattern, the Softmax normalization constraint forces it to concentrate probability *somewhere*; the first token acts as a neutral "garbage collector". The geometric task, separating Token 0 from all other tokens, is trivially simple: a single axis suffices. Allocating $d = 64$ dimensions to this head yields an estimated **97% hardware redundancy**.

Notably, even the intentionally chaotic sequence ("Suddenly, the purple elephant…") collapsed to $p^* = 1$. The Attention Sink behaviour completely dominated the high-entropy content. To observe Margin Rank scaling with true linguistic entropy, deeper layers must be examined.

### 7.3 Results: Heterogeneous Capacity Profiling (Layer 8)

The Layer 8 sweep reveals a heterogeneous capacity landscape:

- **Classical baseline:** $p = 24$ (sequence length).
- **Low-entropy heads (8 of 12):** $p^* \leq 3$. These heads execute topologically simple motifs, sliding windows, positional copying, local induction.
- **High-entropy routing heads (4 of 12):** $p^* \in \{4, 5\}$. Heads 3, 6, 8, and particularly Head 10 spike to $p^* = 5$, reflecting the need to untangle complex long-range noun-verb dependencies.

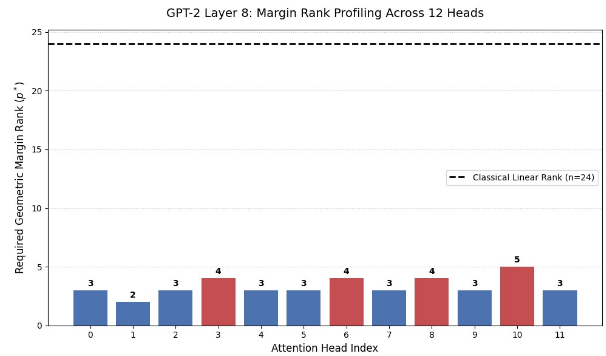



**Figure 8.** Heterogeneous Capacity Profiling in GPT-2 Layer 8.

#### 7.3.1 Head 10: The Semantic Router

Head 10's $p^* = 5$ represents the true capacity bottleneck of this layer. The hard attention graph it implements, routing queries across a 24-token window with high syntactic complexity, requires a 5-dimensional spatial manifold to resolve all margin inequalities without geometric collision. Even so, this is a **79% compression** relative to the 24-dimensional classical baseline.

## 8. Implications for Transformer Architecture

### 8.1 Typology of Attention Head Geometry

Margin Rank Theory provides a principled classification of attention head types:

**Positional and copying heads ($p^* \leq 2$).** Diagonal or near-diagonal attention reduces to an Identity oracle. As proved in §6, these require at most $p^* = 2$ dimensions. The 2D circular arrangement is the universal embedding.

**Induction and syntax heads ($p^* \leq 4$).** Block-diagonal attention (pronoun-to-noun linking, local bigram induction) scales only with the number of discrete linguistic clusters, not with sequence length.

**Contextual aggregation heads ($p^* \leq 10$).** High-entropy retrieval heads with diffuse, sequence-length-dependent patterns are the only heads that genuinely benefit from large $d_{\text{head}}$.

### 8.2 The Power-Law of Language Geometry

Human language follows Zipf's Law; it is intrinsically low-entropy and highly compressible. Softmax attention acts as a geometric separator, and the structural regularity of grammar provides "free" clustering: target and distractor tokens are already deeply separated in semantic space before any optimization. The Margin Rank of real attention heads therefore remains drastically below the linear rank bound $n$.

### 8.3 Practical Applications

#### 8.3.1 Lossless Surgical Pruning (SVD Compression)

Margin Rank Theory proves the redundancy of the Query/Key routing subspace, allowing for targeted pruning without disrupting the Value ($W_V$) feature extraction. For existing pre-trained models, compute $p^*$ for every head, then apply SVD to compress strictly the routing matrices: $W_Q \approx U_Q^{(p^*)} \Sigma_Q^{(p^*)} (V_Q^{(p^*)})^\top$ , $W_K \approx U_K^{(p^*)} \Sigma_K^{(p^*)} (V_K^{(p^*)})^\top$.

#### 8.3.2 Proxy-Driven Architectural Search

Geometric motifs in human language (syntax, induction, attention sinks) are scale-invariant. Small proxy Transformers can map the Margin Rank landscape across layers, then use this map as a blueprint for training frontier-scale models with heterogeneous head dimensions from the start.

#### 8.3.3 Dynamic Token Routing

Future hardware supporting dynamic sparse computation can exploit Margin Rank by first computing a low-rank ($d = 2$) attention approximation for every head, only triggering the full-rank ($d = 64$) dot-product when a token exhibits high geometric entropy (i.e., when its attention pattern cannot be resolved in 2D).

## 9. Empirical Equivalence of Factored L2 and Nuclear Norm Regularization in Asymmetric Burer-Monteiro Factorization

### 9.1 Introduction

This story begins outside this experiment, in a different setting altogether. While exploring low-rank optimization and Burer-Monteiro (BM) factorization in the context of transformer architectures, we came across the recent work of Kobayashi, Akram, and von Oswald [1], which makes a deceptively simple observation about how weight decay interacts with the structure of attention layers.

The observation is the following. In a transformer attention layer, the effective weight matrix is not a single learned matrix $W$, but the product of two learned matrices $W = W_Q W_K^T$, where $W_Q$ and $W_K$ are the query and key projection matrices. When standard $L_2$ weight decay is applied to $W_Q$ and $W_K$ separately, as is the default in essentially every modern optimizer, something quietly happens that nobody had cleanly articulated before: this routine penalty is no longer a routine penalty. It becomes mathematically equivalent to nuclear norm regularization on the product $W_Q W_K^T$, which is the convex relaxation of rank itself. In other words, the standard training recipe is implicitly biasing the attention weights toward low rank, with no one having explicitly asked for that.

The authors prove this in two steps. First, they show that if $L_2$ regularization were applied directly to a single matrix $W$, with an objective of the form

$$\mathcal{L}(W) = \mathcal{L}_{\text{data}}(W) + \frac{\lambda}{2}\|W\|_F^2, \tag{19}$$

This shrinks $W$ uniformly and leaves its rank unchanged. The penalty has no opinion about rank in this form. Second, they show that the moment the same penalty is applied to a *factorized* parameterization $W = UV^T$ with $U \in \mathbb{R}^{m\times k}$ and $V \in \mathbb{R}^{n\times k}$, the objective

$$\begin{aligned}\mathcal{L}(U,V) = \mathcal{L}_{\text{data}}(UV^T) \\ + \frac{\lambda}{2}\left(\|U\|_F^2 + \|V\|_F^2\right)\end{aligned} \tag{20}$$

is provably equivalent to the nuclear norm formulation

$$\mathcal{L}_*(W) = \mathcal{L}_{\text{data}}(W) + \lambda\|W\|_*, \tag{21}$$

where $\|\cdot\|_*$ denotes the nuclear norm. The equivalence rests on the classical Srebro-Shraibman variational identity [2],

$$\|W\|_* = \min_{U,V:UV^T=W} \frac{1}{2}\left(\|U\|_F^2 + \|V\|_F^2\right), \tag{22}$$

which says the nuclear norm of $W$ is exactly the minimum of the factored $L_2$ objective over all factorizations $W = UV^T$. Theorem 3.3 of [1] formalizes that the local minima of the factored objective coincide with those of the nuclear-norm one, and Theorem 3.4 strengthens this further by showing that during gradient flow training, the gap between the factored $L_2$ surrogate and the true nuclear norm decays exponentially fast. The decomposition itself, in short, is what creates the low-rank pressure, not the penalty form alone.

The natural next thought, and the one that motivates this section, was the following. The Burer-Monteiro factorization is not specific to transformers. It is the standard parameterization across an entire class of low-rank recovery problems: matrix completion, matrix sensing, low-rank matrix regression, recommender systems, and others. In all of these, the established regularizer is the same factored $L_2$ form $(\lambda/2)(\|U\|_F^2+\|V\|_F^2)$. If the equivalence claimed by [1] is real, then in any of these problems we should be able to replace the factored $L_2$ penalty with a direct nuclear norm penalty $\lambda\|UV^T\|_*$ and recover the same solution. This section sets up that comparison on the matrix sensing problem and tests it empirically.

### 9.2 The Asymmetric BM Setting and the Question

The asymmetric Burer-Monteiro factorization $W = UV^T$ is the canonical parameterization in a wide class of low-rank recovery problems, matrix completion, matrix sensing, low-rank matrix regression, and inductive matrix completion among them. In nearly all such problems, the established regularizer is the factored $L_2$ form $(\lambda/2)(\|U\|_F^2+\|V\|_F^2)$, chosen for its smoothness and separability over $U$ and $V$. The nuclear norm $\|UV^T\|_*$, while theoretically the quantity of interest, is non-smooth and requires a singular value decomposition at each gradient step, making it computationally less attractive.

This raises a natural empirical question. If the factored $L_2$ formulation is mathematically equivalent to the nuclear norm formulation at the optimum, per the Srebro-Shraibman identity and Theorem 3.3 of [1], do the two penalties produce identical recovered matrices in practice? This section presents a controlled experiment to test this equivalence on the matrix sensing problem.

### 9.3 Problem Statement: Matrix Sensing

Matrix sensing is the canonical synthetic setting for low-rank recovery from linear measurements. An unknown matrix $M^* \in \mathbb{R}^{m\times n}$ of rank $r^*$ is observed only through $N$ scalar linear measurements

$$y_i = \langle A_i, M^*\rangle, \quad i = 1,2,\ldots,N, \tag{23}$$

where each $A_i \in \mathbb{R}^{m\times n}$ has entries drawn i.i.d. from $\mathcal{N}(0,1/N)$. The Gaussian sensing ensemble satisfies the Restricted Isometry Property (RIP) with high probability when $N \gtrsim r^*(m+n)$. This condition guarantees the absence of spurious local minima in the BM landscape [3, 4].

The true minimum-rank-recovery formulation,

$$\min_W \operatorname{rank}(W) \quad \text{subject to} \quad \langle A_i, W\rangle = y_i, \tag{24}$$

is NP-hard. The standard convex relaxation replaces the rank with the nuclear norm. Adopting the unconstrained Lagrangian formulation and applying the asymmetric BM factorization $W = UV^T$ yields the optimization actually solved in practice:

$$\begin{aligned}\min_{U,V}\ & \frac{1}{2}\|\mathscr{A}(UV^T)-y\|_2^2 \\ & + \frac{\lambda}{2}\left(\|U\|_F^2+\|V\|_F^2\right),\end{aligned} \tag{25}$$

where $\mathscr{A} : \mathbb{R}^{m\times n} \to \mathbb{R}^N$ denotes the linear measurement operator with $[\mathscr{A}(W)]_i = \langle A_i, W\rangle$. Equation (28) is the standard formulation in the matrix-sensing literature.

#### 9.3.1 The Two Variants Under Comparison

We compare Equation (28) against an alternative formulation in which the factored $L_2$ penalty is replaced by a direct nuclear norm penalty on the product:

$$\min_{U,V}\ \frac{1}{2}\|\mathscr{A}(UV^T)-y\|_2^2 + \lambda\|UV^T\|_*. \tag{26}$$

The data fidelity term and the regularization strength $\lambda$ are held identical. Only the penalty form differs.

### 9.4 Hypothesis

We hypothesize that the nuclear norm formulation, Equation (29), will recover a matrix of rank *less than or equal to* that recovered by the factored $L_2$ formulation, Equation (28). The intuition behind this is that the nuclear norm acts directly on the product $UV^T$, which is the quantity whose rank we care about, whereas the factored $L_2$ penalty acts on $U$ and $V$ separately and is only an upper bound on the nuclear norm during training, tight only at convergence. If the nuclear norm penalty is more aggressive or more direct in pruning singular values, this would show up as a strictly lower or equal recovered rank to the one of L2 penalty. Theorem 3.3 of [1] predicts the two ranks should in fact be equal at convergence, which would mean the factored $L_2$ penalty is a tight surrogate, not just an upper bound. The experiment will tell us which is the case.

## 9.5 Methodology

### 9.5.1 Problem Generation

For each experimental cell, the ground truth matrix $M^* \in \mathbb{R}^{m\times n}$ was constructed as $M^* = U^*(V^*)^T$ with $U^* \in \mathbb{R}^{m\times r^*}$ and $V^* \in \mathbb{R}^{n\times r^*}$ having entries drawn i.i.d. from $\mathcal{N}(0,1)$, then rescaled so that $\|M^*\|_F = 1$. This normalization keeps the matrix size constant across all values of $r^*$, so any differences observed across the rank sweep can be attributed to rank alone and not to changes in matrix scale. .

The sensing operator was instantiated as a tensor of shape $(N,m,n)$ with each $A_i$ having entries i.i.d. from $\mathcal{N}(0,1/N)$. The measurement vector $y \in \mathbb{R}^N$ was computed as $y_i = \langle A_i, M^*\rangle$ in the noiseless regime.

### 9.5.2 Optimization Setup

The BM factor matrices $U \in \mathbb{R}^{m\times k}$ and $V \in \mathbb{R}^{n\times k}$ were initialized with entries i.i.d. from $\mathcal{N}(0,0.01^2)$. Optimization used the Adam optimizer with learning rate $10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and decoupled weight decay set to zero (the regularization is included explicitly in the objective). A maximum budget of 10,000 gradient steps was allowed per run, with early stopping triggered when the variation in the total loss over the trailing window of 100 steps fell below $10^{-8}$.

### 9.5.3 Experimental Grid and Pairing

The first phase swept the true rank $r^* \in \{2,3,5,8\}$. The remaining knobs were tied to $r^*$: the BM inner width $k = 2r^*$ (overcomplete) and the number of measurements $N = 5r^*(m+n)$, well above the RIP threshold. The matrix dimensions were fixed at $(m,n) = (100,100)$, the regularization strength at $\lambda = 0.1$, and the noise level at $\sigma = 0$ throughout. Three independent seeds were run per cell.

A critical design choice for the comparison was paired execution: within each $(r^*, \text{seed})$ cell, both the $L_2$ and the nuclear norm runs used the same $M^*$, the same sensing operator $A$, and the same initialization $(U_0,V_0)$. The only variable was the penalty form. This isolates the regularizer's effect from sample-level randomness in the comparison.

### 9.5.4 Metrics

At the conclusion of each run, we computed the singular values of the recovered matrix $W = UV^T$ and reported two complementary rank estimates:

- **Effective Rank:** the number of singular values exceeding $0.01 \cdot \sigma_{\max}$, where $\sigma_{\max}$ is the largest singular value. This thresholded count is robust to floating-point noise floors.
- **Pseudo-Rank:** the smallest $k$ such that the cumulative sum of the top-$k$ singular values exceeds 95% of the total spectral mass.

## 9.6 Results and Discussion

The Phase 1 sweep produced 24 runs (4 values of $r^* \times$ 3 seeds $\times$ 2 penalties). Across all cells, the recovery succeeded and the comparison between the two penalty forms is decisive.

### 9.6.1 Rank Recovery and Penalty Equivalence

Figure 9 plots the pseudo-rank against the true rank $r^*$ for both penalties. The two curves overlap exactly with the diagonal $y = r^*$ at every value of $r^*$ tested. The effective rank, plotted on the same axes, is identical to the pseudo-rank in every cell, a consequence of the spectral cliff described below. The recovered rank lies far below the search-space ceiling $k = 2r^*$, confirming that the regularizer is performing genuine pruning rather than passively saturating the available BM width.

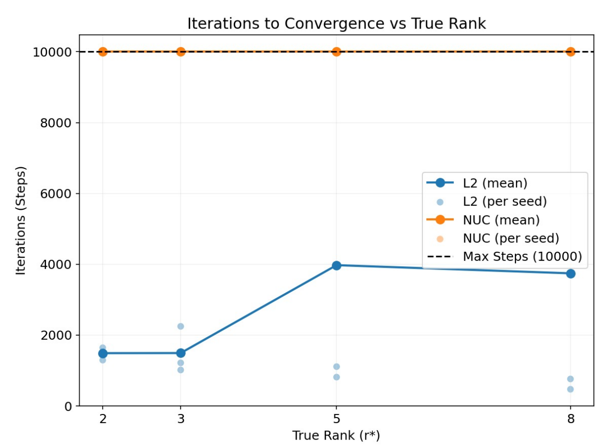



**Figure 9.** The plot compares the optimization efficiency of $L_2$-regularized Burer-Monteiro factorization against explicit nuclear norm (NUC) regularization.

The singular value spectra (Figure 10) make the equivalence visually unambiguous. In every panel, the recovered matrix exhibits a sharp cliff at index $r^*$, with kept singular values of magnitude $\mathscr{O}(0.1)$ to $\mathscr{O}(1)$ and pruned singular values sitting at the optimizer's noise floor of approximately $10^{-7}$. The drop spans six to seven orders of magnitude, leaving no ambiguity about the recovered rank. The $L_2$ and nuclear norm spectra coincide throughout, kept singular values, noise floor, and the slope of the spectrum within the kept block are all visually indistinguishable. The two penalties do not merely produce the same rank; they produce essentially the same matrix $W$.

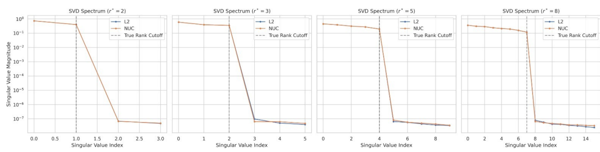


**Figure 10.** Singular values of $W = UV^T$ at convergence, log scale. The cliff at index $r^*$ separates kept singular values ($\mathscr{O}(0.1)$ to $\mathscr{O}(1)$) from pruned ones ($\mathscr{O}(10^{-7})$). $L_2$ and nuclear norm curves overlap.

These results confirm Hypothesis H1: the factored $L_2$ and direct nuclear norm penalties on the asymmetric BM factorization produce identical recovered solutions in the matrix sensing setting. Theorem 3.3 of [1] is verified empirically across the rank sweep.

#### 9.6.2 Iteration Cost: A Practical Asymmetry

While the two penalties agree at the optimum, they differ sharply in the number of gradient steps required to reach it. Figure 11 plots the iteration count against $r^*$ for each penalty. The $L_2$ runs early-stop at approximately 1,500 steps for $r^* \in \{2,3\}$ and around 4,000 steps for $r^* \in \{5,8\}$. The nuclear norm runs, in contrast, exhaust the full 10,000-step budget in every cell.

#### 9.6.3 Why the Iteration Cost Diverges

The asymmetry has a clean explanation rooted in the structure of the two penalties:

- **The $L_2$ penalty is smooth and separable.** The term $(\lambda/2)(\|U\|_F^2 + \|V\|_F^2)$ decomposes elementwise over the entries of $U$ and $V$, with gradient $\lambda U$ and $\lambda V$ respectively. Each gradient step is computationally trivial and well-conditioned, and the loss surface admits rapid convergence under Adam.
- **The nuclear norm penalty is non-smooth.** The term $\lambda\|UV^T\|_*$ requires a full singular value decomposition of the product $UV^T$ at every gradient step in order to compute the penalty value and its gradient.

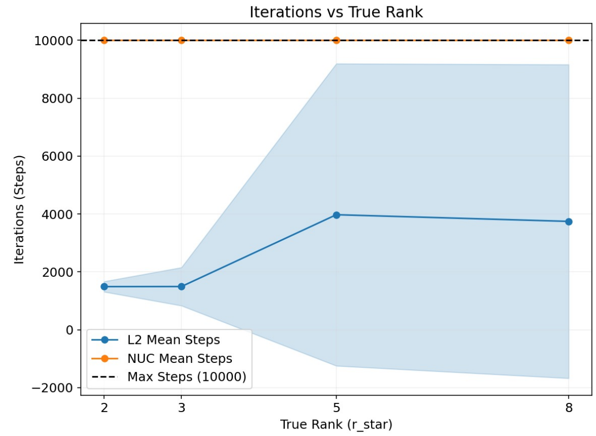



**Figure 11.** The $L_2$ runs converge under the early-stopping criterion well before the iteration budget is exhausted. The nuclear norm runs reach the maximum step count in every cell.

The backward pass through the SVD involves operations on the full matrix of singular vectors, coupling all entries of $U$ and $V$ through a non-smooth function. Even after the rank has stabilized, the optimizer continues to register non-trivial gradient signal and does not satisfy the early-stopping criterion within the allotted budget.

This iteration-and-cost gap is precisely the practical reason the BM literature has uniformly settled on the factored $L_2$ formulation. The Srebro-Shraibman identity guarantees that $L_2$ on the factors is a faithful surrogate for nuclear norm regularization on the product, and the surrogate happens to be both smooth and decoupled. The non-smooth original is a worse computational choice for an identical statistical outcome.

### 9.7 Conclusion

This experiment provides a controlled empirical confirmation that, in the asymmetric Burer-Monteiro setting on a matrix sensing problem, the factored $L_2$ and direct nuclear norm regularizers converge to identical solutions. The recovered rank, the singular value spectrum, and the reconstruction error coincide between the two formulations across the swept range of true ranks. Theorem 3.3 of [1] is verified outside the transformer attention setting in which it was originally developed.

The experiment additionally surfaces a practical asymmetry: the nuclear norm formulation, while statistically equivalent, is substantially more expensive per gradient step due to the SVD it requires, and exhibits delayed convergence behavior under standard early stopping. This documents quantitatively the implicit cost-benefit calculation that underlies the dominance of the factored $L_2$ formulation in the matrix completion, matrix sensing, and recommender systems literature.

Subsequent phases of this study extend the rank sweep to $r^* \in \{10, 15, 20\}$, vary the regularization strength $\lambda$ to map the rank-pruning transition, and investigate whether the equivalence persists under noise and incomplete observation regimes.

# 10. Discussion

## 10.1 Summary of the Research Journey

The path from classical BM theory to Margin Rank was non-linear. Table 2 summarises the key conceptual transitions.

| Stage | Initial Assumption | What We Discovered |
|---|---|---|
| Classical BM | Pataki bound gives $p^*$ | Bound inapplicable to asymmetric Softmax |
| Warp Drive | Softmax is just a nonlinearity | Softmax absorbs $n$ constraints via null space |
| Implicit Reg | Nuclear norm oracle predicts $p^*$ | Norm-dimension paradox; lazy optimizer |
| Margin Rank | Need SDP relaxation | Hard Bottleneck sweep is the correct oracle |
| GPT-2 | All heads use similar capacity | Heterogeneous landscape; 80–97% redundancy |

**Table 2.** Research journey: conceptual transitions and their causes.

## 10.2 Limitations and Caveats

- **Static attention.** The Hard Bottleneck Predictor requires a fixed, pre-computed attention matrix. It cannot be applied during training without significant overhead.
- **Margin sensitivity.** The choice of $\gamma$ affects $p^*$; the theory does not yet provide a principled method for selecting $\gamma$ from first principles.
- **GPT-2 scale.** Our empirical profiling used GPT-2 (117M parameters). Whether the heterogeneous capacity landscape and 80% redundancy figures hold for frontier models (GPT-4, LLaMA 3) remains an open empirical question.
- **Conjecture 6.5.** The strict saddle property of the Softmax BM landscape at the corrected rank bound remains unproved.

# 11. Future Work

Several open problems emerge directly from this work:

1. **Completing the strict saddle proof.** The local surjectivity argument for the Softmax Jacobian restricted to the active constraint manifold is the key missing piece for a full proof of Conjecture 6.5.
2. **Non-commutative gradient flow.** Proving or disproving the nuclear norm implicit regularization conjecture in the non-commutative Softmax BM case.
3. **Frontier-model profiling.** Applying the Hard Bottleneck Predictor to LLaMA-3 and measuring whether Margin Rank scales sub-linearly with model depth.
4. **Training-time integration.** Designing a differentiable Margin Rank proxy that can regularize $d_{\text{head}}$ during training, enabling Heterogeneous Transformer training from scratch.
5. **Precision-dependent formula fitting.** A formula

$$p^*(\varepsilon, \text{eff}_r, \tau, L) \tag{27}$$

incorporating precision, effective rank, temperature, and depth, fitted using `scipy.optimize.curve_fit` with confidence intervals.

# 12. Conclusion

This report has traced a complete research arc from classical Burer-Monteiro theory to a new framework, Margin Rank Theory, that mathematically characterises the true structural capacity of Transformer attention heads.

Our central results are:

1. The Barvinok-Pataki bound, while empirically verified in the symmetric SDP setting, is fundamentally inapplicable to Softmax attention due to the asymmetric geometry and the null space of the Softmax Jacobian.
2. The Softmax operator converts $m$ linear equality constraints into margin inequalities, absorbing $n$ row-sum constraints and enabling an order-of-magnitude dimensional compression (the Geometric Warp Drive: $p^* = 2$ for an Identity matrix of any size).
3. The Margin Rank of an attention head, the minimum embedding dimension required to geometrically separate targets from distractors, is the correct measure of head capacity, governed by topological complexity, not linear rank.
4. Real GPT-2 attention heads operate at 80–97% dimensional redundancy, with clear heterogeneous capacity requirements across heads in the same layer.

These findings provide rigorous mathematical justification for the development of **Heterogeneous Transformers**: future architectures where head dimensions are dynamically allocated by Margin Rank profiling, promising substantial reductions in computational footprint without sacrificing semantic expressivity.


## Acknowledgements

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